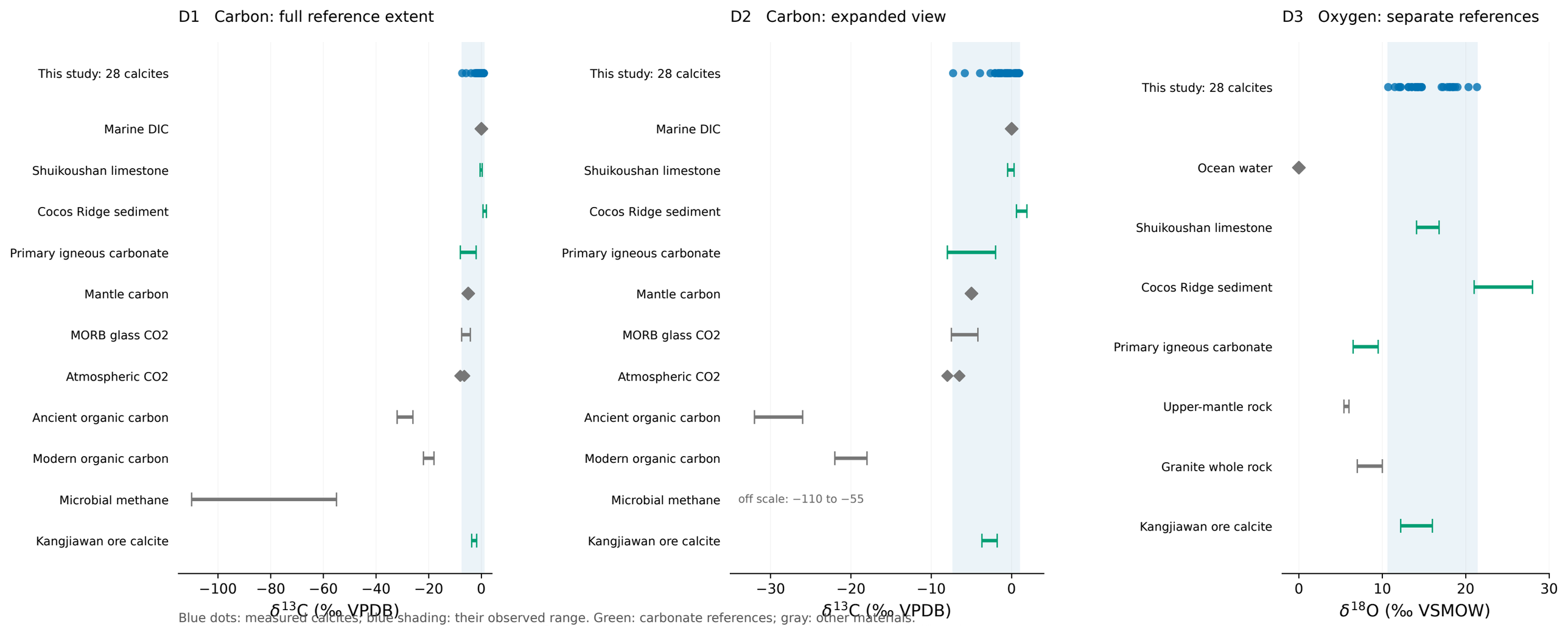

Blue dots: measured calcites; blue shading: their observed range. Green: carbonate references; gray: other materials.
C and O rows are independent comparisons, not paired C–O endmember points. Atmospheric values are two time-specific points.
Water, gas, organic matter and whole-rock references require phase/fractionation constraints before source interpretation.